Inter-annotator reliability and label-quality assessment

5,140 paired annotations • 6 observed categories • 10,000 paired-item bootstrap replicates

MULTICLASS COHEN'S κ

0.786

95% CI 0.772–0.799

OBSERVED AGREEMENT

84.3%

95% CI 83.2%–85.3%

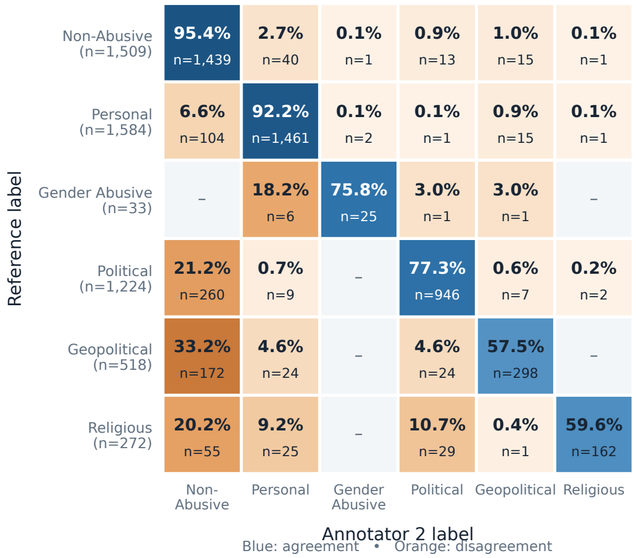
ABUSIVE vs. NON-ABUSIVE κ

0.718

95% CI 0.698–0.738

a Multiclass agreement structure

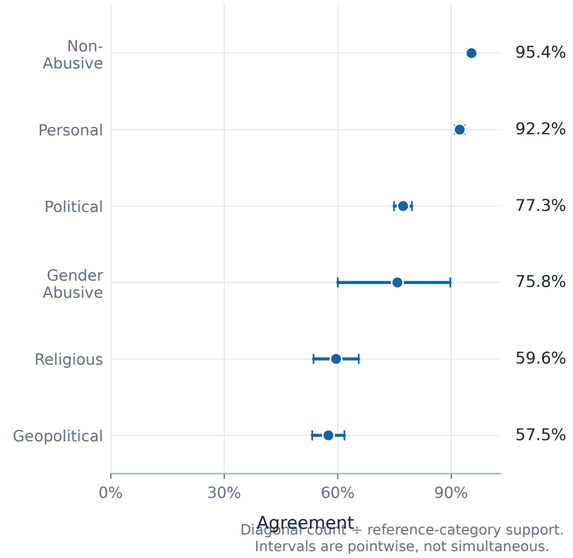
Row-normalized percentages and paired-item counts



Blue: agreement • Orange: disagreement

b Within-category agreement

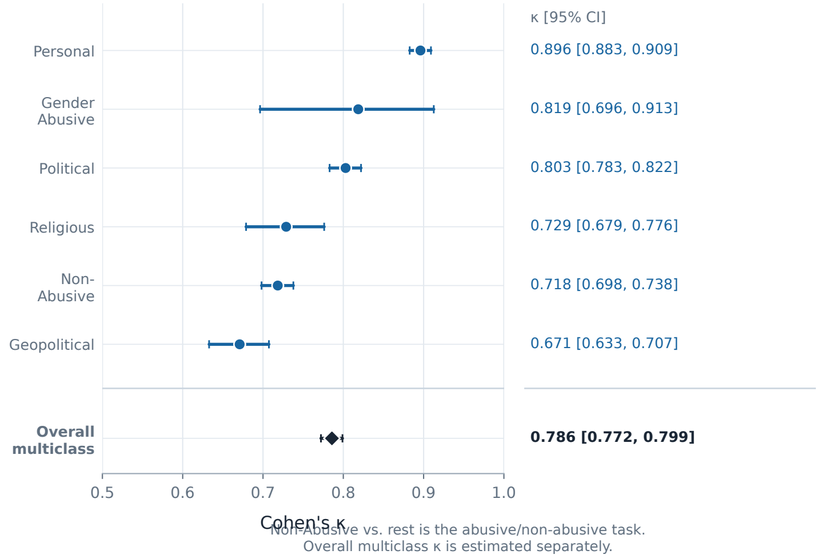
Conditional on the reference category; 95% bootstrap CI



Diagonal count ÷ reference-category support.
Intervals are pointwise, not simultaneous.

c Chance-corrected reliability

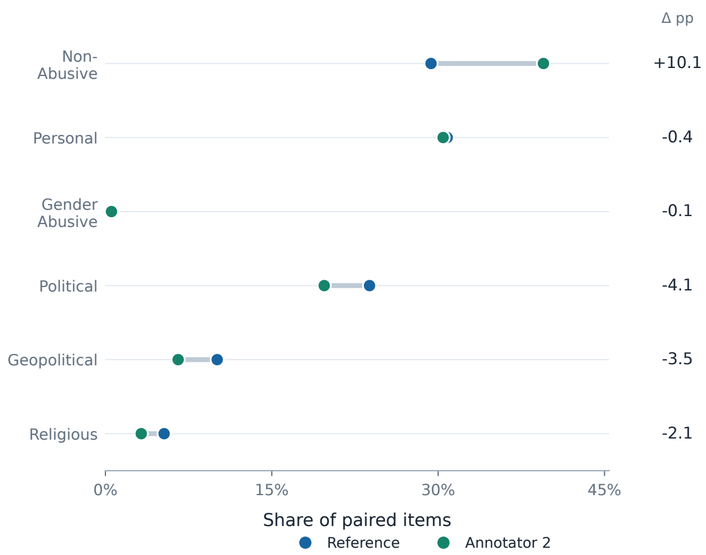
One-vs-rest category κ and overall multiclass κ



Non-Abusive vs. rest is the abusive/non-abusive task.
Overall multiclass κ is estimated separately.

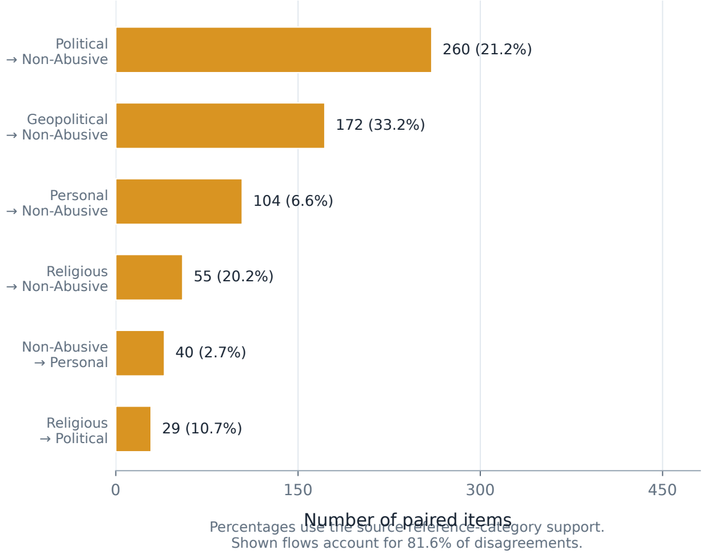
d Annotator label distributions

Right column: Annotator 2 − reference, in percentage points



e Largest directed disagreements

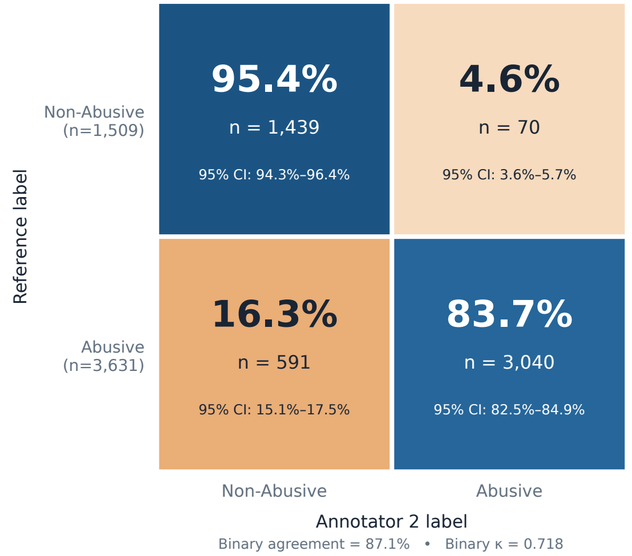
809 discordant annotation pairs



Percentages use the source/reference-category support.
Shown flows account for 81.6% of disagreements.

f Abusive / non-abusive agreement

All abusive subtypes collapsed into a single category



Binary agreement = 87.1% • Binary κ = 0.718

Intervals: pointwise 95% percentile bootstrap CIs from 10,000 paired-item resamples; undefined replicate estimates are excluded per metric.

Reference denotes the first annotation column, not necessarily ground truth. Row-normalized percentages condition on reference labels; pp denotes percentage points.